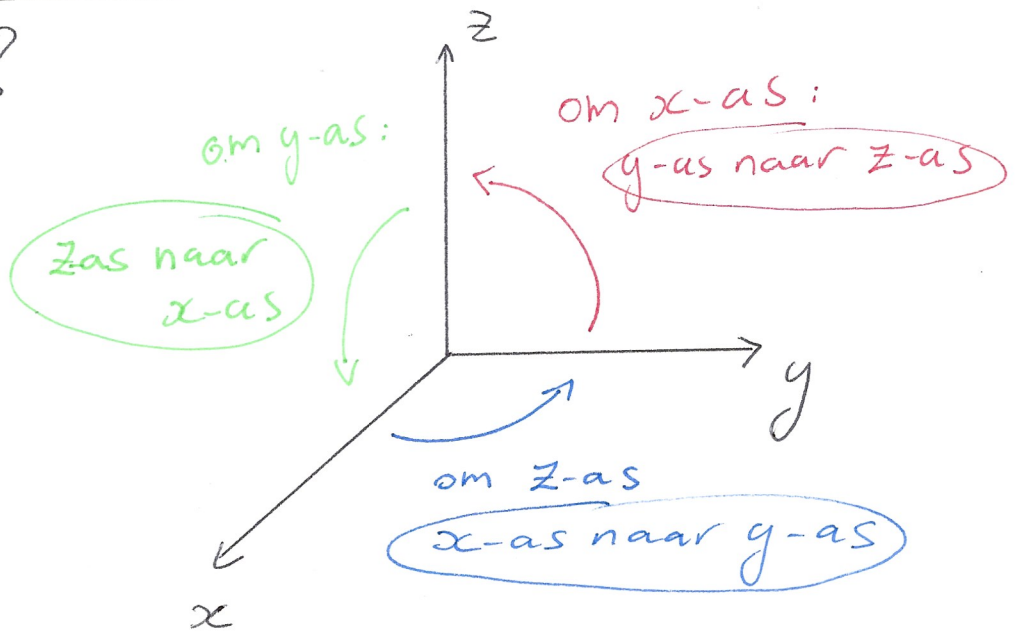


Rotates in 3D: on x, y & z -as

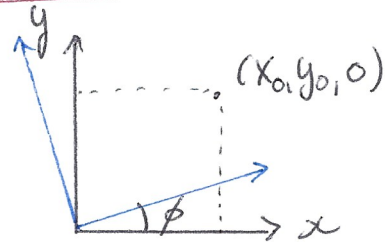
rotationsrichtung?
rechterhand regel



Punt $P(x_0, y_0, z_0)$: nieuwe coördinaten (x_1, y_1, z_1) ?

$\left\{ \begin{array}{ll} \text{rotatie rond } z\text{-as:} & \underline{z\text{-coördinaat blijft zelfde}} \\ \text{" " } y\text{-as:} & y\text{-coördinaat " " } \\ \text{" " } x\text{-as:} & \underline{x\text{-coördinaat " "}} \end{array} \right.$

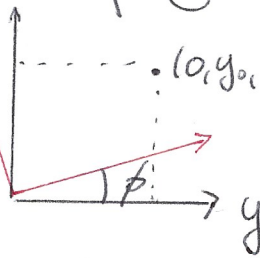
Rond z-as: xy -vlak



$(x_1, y_1, z_1) = (x_0, y_0, z_0)$

$$\begin{pmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
 2D-rotate matrix

Rond x-as: yz-vlak:



$(x_1, y_1, z_1) = (x_0, y_0, z_0) \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi & \cos \phi \end{pmatrix}$

2D rotate matrix